

# TEJAS RAMAKRISHNAN

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## EXPERIENCE

Samsung Research

Engineer- Reality Media Lab

Seoul

Sep 2024 – Present

- Built on-device **Gaussian Splatting** based **3D reconstruction pipeline**
- Utilized **VI-SLAM**, **stereo-depth**, **threading** and **GPGPU programming** to achieve a PSNR of **27dB+** with **<5 minute** end-to-end latency.
- Invented **6DoF Video Stabilization** system, registering **A-grade patent**
- Used a pipeline of transformer based **pose-estimation**, **6DoF smoothing**, and **frame inpainting** with **diffusion**(TrajectoryCrafter)
- Created real-time face-detection based object centering VDIS system

Samsung Research

Research Intern- Augmented Reality Lab

Seoul

May 2023 – Jul 2023

- Developed a **generative pipeline** to synthesize 3D human-motion datasets for **6DOF** motion prediction and AR latency reduction.
- Integrated **motion diffusion** with **autoregressive scene completion** and probabilistic sampling to create **diverse 3D environments**

SURGE, IIT Kanpur

Research Intern under Prof. Ashutosh Modi

India

Jan 2022 – Dec 2022

- Built a **3000+ size multimodal emotion dataset** from Indian sources.
- Developed annotation tools by extending **Label Studio**.
- Evaluated **deep learning models** for multimodal emotion recognition.

## PROJECTS

Repairing Neural Networks

2024

- Identified training data responsible for model failures using **linear influence functions** under a **Bayesian continual learning** framework.
- Repaired models using "machine unlearning" with **Newton Update Removal** and **Elastic Weight Consolidation**, avoiding full retraining.

Multiple Object tracking

2023

- Implemented a lightweight **MOT system** using **YOLOv5 + DeepSORT**
- Fine-tuned YOLO on **MOT16 dataset**, accounting for class imbalance
- Improved track stability by introducing **geometric motion penalties**

Java Compiler

2023

- Built an end-to-end **Java-to-x86 compiler** (**10K+ LOC**) in C++, including an LR(1) parser, AST construction, and symbol tables.
- Engineered an optimized pipeline using 3AC with **dead code elimination** and **jump optimization** to enhance efficiency.

CAPTCHA Recognition

2022

- Built a **CNN-based sequence prediction model** to decode CAPTCHA images containing distorted alphanumeric characters.
- Achieved **99% decoding accuracy** on the evaluation dataset.

## TECHNICAL SKILLS

3D Reconstruction

Video Stabilization

C/C++

OpenCL

Python

PyTorch

SQL

Git

Bash

OpenCV

## EDUCATION

B.Tech in Computer Science

Indian Institute of Technology Kanpur

Aug 2020 – May 2024

GPA: 9.7/10.0

CBSE(XII)

May 2020

Grade: 98.2%

## COURSEWORK

- Machine Learning
- Deep Learning for Computer Vision
- Probabilistic Machine Learning
- Reinforcement Learning
- Data Structures & Algorithms
- Operating Systems
- Compiler Design

## ACHIEVEMENTS

- Ratan Swarup Memorial Prize** – Best All-Round Graduating Student, IIT Kanpur
- Academic Excellence Award** – Top 10% for three consecutive years
- AIR 201** in JEE Advanced 2020
- AIR 272** in JEE Main 2020
- AIR 350 KVPY 2020**
- Secured exceptional **A\* grade** in DSA, Machine Learning, Discrete Structures, and Reinforcement Learning
- Achieved a **perfect SPI of 10** in four out of eight semesters

## OTHER

- Music Club Secretary – Media & Cultural Council, IIT K, Trained in **Classical Music**
- Represented IIT Kanpur at the **55th Inter-IIT Sports Meet** in football
- Tutored underprivileged students at **Prayas, IIT-Kanpur**
- Semester Exchange – Hong Kong Baptist University